Printables by JOSEF PRUSA



Open-Source Syringe Pumps

LIBRE hub

VIEW IN BROWSER

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Summary

3D printable syringe pumps for laboratory or hobby use, including microfluidics, mass-spectrometry, and microscopy.

Learning > Chemistry & Biology

Tags: motor stepper electronics pump microfluidics opensource syringe m3 openhardware droplet heatinsert librehub

The open-source syringe pump is a standard research instrument for use in microfluidics and several other laboratory workflows. There are numerous designs available in the realm of open-source syringe pumps. We have combined some the best features available based on a pre-existing versatile design.

To build these pumps you can check our free **documentation**. To make modifications see our **online CAD model**.

Our **strobe-enhanced microscopy stage** works great with these syringe pumps, to complete an open-source workstation for imaging fast processes, particularly developed for microfluidic droplet generation.

This is a re-design and modification of the "Open-Source Syringe Pump" (documentation) developed by Andrey Samokhin, made by Matías Hurtado Labarca and Tobias Wenzel as part of the **LIBRE hub** project.

This remix is based on

Open-Source Syringe Pump	**Open-Source Syringe Pump**

Model files

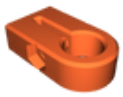


assembled-syringe-pump.stl

Simulation



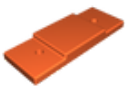
small-syringe-holder.stl



add-on-end-stop-limit.stl



add-on-thread-holder.stl

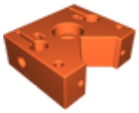


large-syringe-holder.stl



add-ons-to-operate-pumps-in-retraction-mode.stl

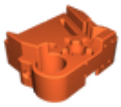
Optional Holding System for retraction



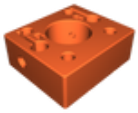
base-syringe-holder.stl



knob.stl



base-carriage.stl



base-stepper-motor-plate.stl

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